

has support for WKB and WKT formats. The VirtualNetwork module enables Spatialite to handle routing through the implementation of the Dijkstra's Shortest Path algorithm. Spatial indexing is available through SQLite's RTree extension.

Spatial file conversion

Finding the spatial data you need is a project in itself. Furthermore, you will often find that your GIS won't be able to handle the file formats your spatial data is available in, considering the number of different file types. For example, OSM XML (openstreetmap XML format) won't load in uDig, and Google Earth won't read Shapefiles.

ogr2ogr

ODR is a set of command-line utilities to read and write spatial vector files. Ogr2ogr, which is a part of the ODR set, is a utility that converts from one spatial file format to another. During the conversion, it allows operations like attribute selections, resetting of the output coordinate system or even reprojection of the features.

This utility can be used to merge several spatial files into a single file with the help of the -append flag. Ogr2ogr can work with the following file formats: shapefile, GML, KML, TIGER, Mapinfo and the PostGIS database, among others.

shp2pgsql

This command line utility converts a Shapefile into SQL, which can later be loaded into a PostGIS-enabled Postgres database. It has flags to create tables, append to existing tables, output in the WKT format, etc.

osm2pgsql

This is a command-line utility used to directly load an Openstreetmap XML file into a PostGIS database. It creates all the tables itself. Once the data is in the database, ogr2ogr could be used to export it in other formats.

In certain cases, you may have to do a multi-tier conversion—that is, format A -> B, then B->C. uDig and QGIS read a number of formats and allow you to save in other formats, and hence serve as file conversion tools. 

Glossary

- DRG - Digital raster graphic
- GML - Geography Markup Language
- JGrass - Java Geographic Resources Analysis Support System
- KML - Keyhole Markup Language
- RCP - Rich Client Platform
- TIGER - Topologically Integrated Geographic Encoding and Referencing
- UDIG - User friendly Desktop Internet GIS
- WKB - Well Known Binary
- WKT - Well Known Text
- WMS - Web Map Service

References

- ogr2ogr: <http://www.gdal.org/ogr/>
- OpenLayers: <http://openlayers.org/>
- osm2pgsql: <http://wiki.openstreetmap.org/wiki/Osm2pgsql>
- pgRouting: <http://pgrouting.postgis.net/>
- PostGIS: <http://postgis.refractions.net/>
- QGIS: <http://www.qgis.org/>
- Spatialite: <http://www.gaia-gis.it/spatialite/>
- uDig: <http://udig.refractions.net/>

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